

Problem-Solution Fit

Date	5 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] • Loan applicants (salaried & self-employed individuals) seeking home, personal, or business loans • Bank loan officers & credit analysts who evaluate applications • Financial institutions & NBFCs processing high volumes of loan requests • Students & researchers studying ML-based credit risk systems	6. CUSTOMER LIMITATIONS [CL] • Limited financial literacy — applicants don't know what factors affect approval • No access to credit scoring tools • Low digital literacy among rural applicants • Banks lack automated, explainable AI tools for quick decisions	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Manual Bank Review — Pro: Personalized; Con: Slow, biased, inconsistent • Credit Score Apps (CIBIL) — Pro: Accurate credit data; Con: Only shows score, no approval prediction • Existing Loan Eligibility Calculators — Pro: Simple; Con: Rule-based, not ML-driven, low accuracy • Smart Lender (Our Solution) — Pro: ML-powered, instant, explainable; Con: Educational-level, no live bank integration yet
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Loan approval process is slow and takes days to weeks — Happens with every application • Applicants are unaware of why their loan was rejected — Very frequent • High rejection rates due to incomplete or poorly understood applications — Frequent • Banks face operational bottlenecks processing hundreds of applications manually — Daily occurrence	9. PROBLEM ROOT / CAUSE [RC] • Banks rely on manual review by loan officers, leading to delays and human bias • No transparent system to explain which factors (credit history, income, etc.) led to rejection • Lack of a pre-screening tool that tells applicants their eligibility before formally applying • No data-driven automation in traditional loan assessment workflows	7. BEHAVIOR + ITS INTENSITY [BE] • Applicants repeatedly visit bank branches or portals to check status — High intensity • Applicants guess and reapply with minor changes, wasting time — Medium intensity • Bank officers spend excessive time on borderline cases instead of clear approvals — High intensity • Applicants feel anxious and uncertain during the waiting period — High emotional intensity

3. TRIGGERS TO ACT [TR]

- Applicant needs urgent funds (medical, business, home purchase)
- Applicant previously rejected without explanation and wants to understand why
- Bank wants to reduce manual processing time and improve decision consistency
- Financial institution adopting digital transformation / automation initiatives
- Student or researcher exploring ML use cases in fintech

4. EMOTIONS BEFORE / AFTER [EM]

BEFORE using Smart Lender: Anxiety, uncertainty, confusion about eligibility criteria, fear of rejection, frustration from slow bank processes, lack of confidence when applying.

AFTER using Smart Lender: Clarity on eligibility status, confidence to proceed or improve application, relief from instant result, better understanding of key factors (credit history, income), empowerment to make informed financial decisions.

10. YOUR SOLUTION [SL]

Smart Lender is an AI-powered web application that predicts loan eligibility instantly using an XGBoost ML model trained on real loan data.

- User fills a simple form with 11 inputs (income, credit history, education, property area, etc.)
- XGBoost model (81.1% test accuracy) processes the inputs in seconds
- Returns Approved or Rejected result with applicant summary and key decision factors (credit history, income, education, property area)
- Allows resubmission so applicants can adjust inputs and re-check eligibility
- Built with Python, Flask, and a clean responsive web UI

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Bank websites & portals
- Loan eligibility apps
- Social media & word of mouth

OFFLINE

- Bank branches & loan advisors
- College placements & financial counselors